

Thermo Scientific™ HyPerforma™ Glass Bioreactor 15 L

thermo-glass-15

Field-level manufacturer evidence; remaining values unverified

Each field is classified independently. Direct manufacturer values take precedence over estimates only for the matching model and configuration. Approximate values printed by the manufacturer remain approximate. Unverified fields are editable simulation assumptions, not equipment operating limits.

Parameter	Value in simulator	Evidence status
Minimum working volume	~6 L	Source-supported
Maximum working volume	14.66 L	Source-supported
Nominal vessel / bag size	15 L	Source-supported
Total vessel volume loaded into the form	16.8 L	Source-supported
Initial working volume loaded into the form	6 L	Derived / estimated
Product-contact material	Borosilicate glass / 316L stainless steel / silicone	Source-supported
Height : diameter	2.2	App assumption
Internal / chamber diameter	0.204 m	Derived / estimated
Liquid height at maximum working volume	0.4487 m	Derived / estimated
Impeller / mixing type	Pitched-blade turbine	App assumption
Impeller count	2	App assumption
Impeller diameter	0.0673 m	Derived / estimated
Power number	1.3	App assumption
Initial agitation	350 rpm	App assumption
Maximum agitation	1250 rpm	Source-supported
Initial aeration	0.3 vvm	App assumption
Maximum aeration	2 vvm	App assumption
Baffled vessel	Yes	App assumption
Maximum pressure	0.5 bar(g)	App assumption
Default operating pressure	0 bar(g)	App assumption
Maximum inlet oxygen fraction	100%	App assumption

Parameter evidence and derivations

Minimum working volume - ~6 L

Source-supported | Thermo HyPerforma glass bioreactor datasheet - PDF page 2; Specifications table, 15 L column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: ~6 L. The manufacturer prints an approximate value (~); this is not an app-derived estimate.

Limitations: Loaded working volume and total loaded volume are distinct from the named vessel size.

Local original: [Thermo HyPerforma glass bioreactor datasheet - PDF page 2](#)

[Manufacturer publication](#)

Maximum working volume - 14.66 L

Source-supported | Thermo HyPerforma glass bioreactor datasheet - PDF page 2; Specifications table, 15 L column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 14.66 L.

Limitations: Loaded working volume and total loaded volume are distinct from the named vessel size.

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Nominal vessel / bag size - 15 L

Source-supported | Thermo HyPerforma glass bioreactor datasheet - PDF page 2; Specifications table, 15 L column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 15 L.

Limitations: Loaded working volume and total loaded volume are distinct from the named vessel size.

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Total vessel volume loaded into the form - 16.8 L

Source-supported | Thermo HyPerforma glass bioreactor datasheet - PDF page 2; Specifications table, 15 L column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 16.8 L.

Limitations: Loaded working volume and total loaded volume are distinct from the named vessel size.

Local original: [Thermo HyPerforma glass bioreactor datasheet - PDF page 2](#)

[Manufacturer publication](#)

Initial working volume loaded into the form - 6 L

Derived / estimated | Form-population rule

Rule: initial working volume = min(minimum working volume, maximum working volume)

Selection / calculation: min(6 L, 14.66 L) = 6 L

Limitations: The preset starts the process at the lower end of its configured working range.

Product-contact material - Borosilicate glass / 316L stainless steel / silicone

Source-supported | Thermo HyPerforma glass bioreactor datasheet - PDF page 2; Specifications table, 15 L column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: Borosilicate glass / 316L stainless steel / silicone.

Limitations: Loaded working volume and total loaded volume are distinct from the named vessel size.

Local original: [Thermo HyPerforma glass bioreactor datasheet - PDF page 2](#)

[Manufacturer publication](#)

Height : diameter - 2.2

App assumption | Generic reusable glass profile

Rule: PROFILE_DEFAULTS["glass"].heightDiameterRatio = 2.2

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable glass starting profile. This ratio is subsequently used to estimate vessel diameter and liquid height.

Internal / chamber diameter - 0.204 m

Derived / estimated | Cylindrical geometry estimate

Rule: $D = \sqrt[3]{(4 \times V)/(\pi \times H:D)}$

Selection / calculation: $D = \sqrt[3]{(4 \times 0.01466 \text{ m}^3)/(\pi \times 2.2)} = 0.2039579 \text{ m}$; the form displays 0.204 m.

Limitations: The estimate treats the liquid volume as a straight-sided cylinder and does not include heads, cones, internals or manufacturer-specific dimensions.

Liquid height at maximum working volume - 0.4487 m

Derived / estimated | Cylindrical geometry estimate

Rule: liquid height = estimated internal diameter x H:D

Selection / calculation: $0.2039579 \text{ m} \times 2.2 = 0.4487073 \text{ m}$; the form displays 0.4487 m.

Limitations: This is linked to the same ideal-cylinder approximation used for the diameter.

Impeller / mixing type - Pitched-blade turbine

App assumption | Generic reusable glass profile

Rule: PROFILE_DEFAULTS["glass"].impellerType = "pitched"

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable glass starting profile. The selected type determines the generic mixing archetype shown and is paired with the preset power-number assumption.

Impeller count - 2

App assumption | Generic reusable glass profile

Rule: PROFILE_DEFAULTS["glass"].impellerCount = 2

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable glass starting profile. The count is used by the vessel diagram and represents a generic configuration unless source-supported.

Impeller diameter - 0.0673 m

Derived / estimated | Generic impeller-to-tank ratio

Rule: impeller diameter = $\max(0.33 \times \text{estimated vessel diameter}, 0.001 \text{ m})$

Selection / calculation: $\max(0.33 \times 0.2039579 \text{ m}, 0.001 \text{ m}) = 0.0673061 \text{ m}$; the form displays 0.0673 m.

Limitations: A one-third tank-diameter impeller is used only as a generic starting geometry when no model-specific impeller diameter is stored.

Power number - 1.3

App assumption | Generic reusable glass profile

Rule: PROFILE_DEFAULTS["glass"].powerNumber = 1.3

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable glass starting profile. The power number is an empirical generic value used in $P = N_p N^3 D_i^5$. It is not recalculated from blade geometry.

Initial agitation - 350 rpm

App assumption | Generic reusable glass profile

Rule: `PROFILE_DEFAULTS["glass"].baseRpm = 350`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable glass starting profile. This is the initial agitation setting loaded into the form.

Maximum agitation - 1250 rpm

Source-supported | Thermo HyPerforma glass bioreactor datasheet - PDF page 2; Agitator speed

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 1250 rpm.

Limitations: Published default configured speed; TruBio can configure lower values. Not a universal process-specific safe speed.

Local original: [Thermo HyPerforma glass bioreactor datasheet - PDF page 2](#)

[Manufacturer publication](#)

Initial aeration - 0.3 vvm

App assumption | Generic reusable glass profile

Rule: `PROFILE_DEFAULTS["glass"].baseVvm = 0.3`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable glass starting profile. This is the initial gas rate loaded into the form.

Maximum aeration - 2 vvm

App assumption | Generic reusable glass profile

Rule: `PROFILE_DEFAULTS["glass"].maxVvm = 2`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable glass starting profile. This is the gas-rate ceiling available to the generic DO-control cascade.

Baffled vessel - Yes

App assumption | Generic reusable glass profile

Rule: `PROFILE_DEFAULTS["glass"].baffled = true`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable glass starting profile. This Boolean changes the generic vessel configuration and mixing interpretation.

Maximum pressure - 0.5 bar(g)

App assumption | Generic reusable glass profile

Rule: `PROFILE_DEFAULTS["glass"].maxPressure = 0.5`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable glass starting profile. This is the pressure ceiling used for warnings in the generic equipment model.

Default operating pressure - 0 bar(g)

App assumption | Generic reusable glass profile

Rule: `PROFILE_DEFAULTS["glass"].operatingPressure = 0`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable glass starting profile. This is the starting headspace pressure loaded into the form.

Maximum inlet oxygen fraction - 100%

App assumption | Fixed simulator capability assumption

Rule: maxOxygenFraction = 100

Selection / calculation: No calculation. Every vessel preset currently permits a 100% oxygen inlet ceiling in the generic gas cascade.

Limitations: This describes the simulator's permitted gas-mixture range, not proof that the selected hardware is certified for pure oxygen service.

Configuration notes and unresolved evidence

2026-09-11 field-level review: use the evidence beside each parameter; an archived document does not verify the whole preset.

Headplate diameter and bioreactor height are external assembly dimensions, not internal diameter or liquid height; they are not used in the geometry model.

No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

Local original document identities

Thermo HyPerforma glass bioreactor datasheet

thermo-hyperforma-glass-bioreactor-datasheet-cf8668407b91.pdf

SHA-256: cf8668407b91ff6cfad274438123ca986e84c3d86b597b6320c0aff7008bea7c